Quadrilateral particle arrangement within shocks in a two-dimensional dusty plasma

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The microscopic structure within a two-dimensional shock was studied using data from a dusty plasma experiment. A single layer of charged microparticles, levitated in a glow-discharge plasma, was perturbed by an electrically floating wire that was moved at a steady supersonic speed to excite a compressional shock. A rearrangement of particles was observed, from a hexagonal lattice in the preshock into a quadrilateral microstructure on the front side of the shock. This quadrilateral structure would not be stable in a monolayer of identical repulsive particles, under equilibrium conditions. Glaser-Clark polygon analysis of the microstructure helped in identifying quadrilaterals. Voronoi analysis was used to characterize the defect fraction behind the shock, as an indication of shock-induced melting.

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I. INTRODUCTION

Shock waves (shocks), which can be defined as nonlinear disturbances propagating at supersonic speed, exist in all kinds of media, including gases, solids, and plasmas. One type of plasma, which can also sustain shocks, is two-dimensional (2D) dusty plasma. This type of plasma consists of a layer of microparticles levitating in the sheath above the lower electrode in the gas discharge [1,2]. The ground state for such a 2D layer is a triangular lattice with sixfold symmetry.

Shocks in 2D dusty plasma have recently seen an increase in interest, including experimental and molecular dynamics (MD) simulation studies [3–20]. Among recently reported MD simulations, we can mention studies of shock-front speed and hydrodynamic quantities [7], hydrodynamic quantities and velocity distribution functions along with analytical expression between the thermal and the drift velocities [10], the evolution of blast waves [15], and a head-on collision of shocks [16].

Due to its unique properties, 2D dusty plasma is an excellent medium for experimentally studying the microscopic structure of shocks. The sound speed in that medium is the order of centimeters per second so that video microscopy diagnostics allow the resolving and tracking of individual microparticles [21]. These particle-level measurements allow experimental study of phenomena such as the finite thickness of the shock front [14], and they also allow direct comparisons to MD simulations. The results may also be applicable to other 2D physical systems that similarly exhibit an equilibrium state with hexagonal structure [20].

The microscopic properties of shocks in 2D dusty plasma have been studied using the method of Voronoi diagrams, with an input of particle-position data obtained from MD simulations. Sun and Feng [12] studied melting, which was reported to occur when the moving exciter had a speed exceeding a threshold, as determined by Voronoi diagrams that were examined qualitatively and were also used to quantify the defect fraction. A leap of the defect ratio, which can indicate the occurrence of a phase transition from the solid to liquid states, was observed. Qiu *et al.* [13] studied a postshock region of 2D crystal, with both liquid and solid coexisting phases, which was observed when the exciter speed was sufficiently high. For this phenomenon, again Voronoi diagrams were used to examine disorder and quantify the defect fraction.

In this paper, we report an experimental investigation of a point that has not been much studied yet: the microscopic structure on each side of the shock. We find that the sixfold triangular geometry of a ground-state crystal does not persist under the sudden compression of a shock, but instead the particles tend to rearrange into a quadrilateral microscopic structure. The quadrilateral microstructure is unusual, because for a 2D crystal with isotropic repulsion the triangular (also called hexagonal) lattice is the only close-packed lattice in two dimensions [22], under the conditions of steady equilibrium. A quadrilateral lattice in 2D dusty plasma under steady conditions can exist only in special cases, such as binary mixtures [23] or when the crystal buckles [24,25]. In this paper, we observe quadrilateral microscopic structure under the conditions of shock compression that are not a steady equilibrium.

To study this microscopic structure, we analyze our experimental particle-position data using the methods of Voronoi diagrams [26] and the polygon construction method of Glaser and Clark [27–29]. The polygon construction method is particularly effective in revealing the quadrilateral microscopic structure that we report, for the nonequilibrium conditions of shock compression in a 2D substance.

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II. METHOD

A. Experiment

In this paper, we report a further analysis of experimental runs that we previously reported Ref. [9]. In that experiment, as an exciter was moved at a steady speed, there was a localized compression of the 2D dust layer in front of the exciter. This compression was strong and supersonic, indicating that it was a shock. The main result in that previous report was the relationship of two speeds, for the shock itself and for the exciter. Here we analyze the particle-position data for a different purpose, to assess the microscopic structure on both sides of the shock. We briefly review that experiment next; further details are provided in Ref. [9].

The experiment was performed in our Kuda-Topf chamber [9,30,31]. To sustain a gas discharge in argon, 13.56-MHz radio-frequency power with a peak-to-peak voltage of 184 V was applied to the lower electrode through a coupling capacitor, while the outer wall of the chamber was grounded. As a result, a dc bias of -85.5 V developed on the lower electrode. A 2D layer of monodisperse (8.69 μm in diameter) melamine-formaldehyde microparticles was levitated in the plasma sheath above the lower electrode. The particle charge was -1.4×10^4 elementary charges. In the absence of the perturbation of a shock, the lattice constant was $b = 0.80$ mm, and correspondingly the 2D Wigner-Seitz radius was $a = 0.42$ mm, while the nominal 2D dust plasma frequency was $\omega_{\rm pd} = 49\ {\rm s}^{-1}$.

Compressional shocks were generated by continuously moving an exciter at a controlled velocity. The exciter was a nickel wire, shown in the Supplemental Material [32]. This exciter wire was aligned perpendicular to its direction of motion but parallel to the 2D microparticle cloud. Runs were performed at five different speeds of the exciter, ranging from 50.8 to 101.6 mm/s all above the sound speed of 19 mm/s. In dimensionless units, this range was from 2.5 to 5.0 $a\omega_{\rm pd}$, significantly exceeding the exciter speeds in the simulation of Sun *et al.* [15].

For this paper, we analyzed the microscopic structure for all of these runs; we report snapshots of the data for three runs in the main text including the unperturbed state before the shock, along with more detailed video data for all the runs in the Supplemental Material [32], including videos of the development of the microscopic structure as the shock passes through the region of interest.

In the experiment, the 2D microparticle cloud was imaged at 870 frames/s using video microscopy [33–35]. Afterwards, positions of individual microparticles were measured in each video frame, using the moment method, which has a subpixel precision [33,36]. We restricted our analysis to particles within a 25.4×10.2 mm region of interest (ROI), as described in the Supplemental Material of Ref. [9]. The microparticle positions within the ROI, in each video frame, served as the input for the analysis that we describe next.

B. Polygon analysis

Two methods of structural analysis were used for this paper: Voronoi analysis [26] and the Glaser-Clark polygon analysis method [28,29]. Both methods have been used in the literature to characterize defects, for the melting of a 2D crystalline lattice. Both of these methods start with particle position measurements as the input for a Delaunay triangulation calculation, which identifies bonds as line segments connecting nearby particles. In an ideal 2D crystal under equilibrium conditions at zero temperature, these bonds will be arranged as equilateral triangles, i.e., with sixfold symmetry. However, in actual conditions, they will not be perfectly equilateral.

The Voronoi method has been used to study 2D melting with experimental data since at least the 1987 colloid experiments of Ref. [37]. The Voronoi algorithm is to form a polygon having its edges being the perpendicular bisectors of the bonds. In an equilibrium state of a 2D crystal, these Voronoi polygons will be hexagons, which is to say the local geometry has sixfold coordination. Five- and seven-sided polygons, and any other polygon that is not sixfold, are identified as a disclination, which is a defect. This kind of defect analysis has an essentially binary outcome: the vertex at the center of a polygon is either sixfold or it is not. Sufficient displacements of particles from the ground-state arrangement can result in a pair of polygons changing from hexagons to a five-seven pair of Voronoi cells. Sometimes only a tiny displacement of a single particle can result in such a change, from nondefect to defect. This binary classification, defect vs nondefect, is a limitation of the Voronoi method.

The Glaser-Clark polygon analysis method, which was developed in 1990 [28], has a distinctive difference compared to the Voronoi approach because it does not have a simple binary outcome of defects vs nondefects. Instead, Glaser-Clark polygons have various orders: triangular, quadrilateral, pentagonal, or even higher-order polygons. A triangular polygon is nondefective. A quadrilateral is the least severe defect, followed by a pentagon and so on for more severe defects [27,38]. This gradation of defects is an advantage of the Glaser-Clark method. The algorithm for producing these polygons from the Delaunay triangulation is to remove bonds if their angles deviate sufficiently from an equilateral shape. The criterion we use, as in Refs. [27–29], is to remove bonds opposite an unusually large angle of 75°. We used the source code for Glaser-Clark polygon analysis provided in the Supplementary Material of Ref. [27]. Despite its advantages, this method has not been used as much as Voronoi analysis in the literature. Glaser and Clark originally demonstrated it using simulation data for a 2D lattice under steady conditions [28,29]; it was then used with experimental data to characterize the sudden melting of a 2D dusty plasma crystal [27,30], the melting process in 2D Yukawa systems [38], and the structure and dynamics in screened 2D dipole systems [39].

III. RESULTS

In Figs. 1–3, we present 2D maps of the microstructure. The particle positions shown in the upper rows of these figures were the input to a Delaunay triangulation calculation, for preparing the Voronoi polygons in the middle rows and the Glaser-Clark polygons in the lower rows. These three figures are for different runs, each with a different speed for the exciter wire, which moved in the positive x direction. In Fig. 1, we also provide an example of the conditions of the microparticle cloud before it was disturbed by the movement of the exciter. In the Supplemental Material [32], we provide

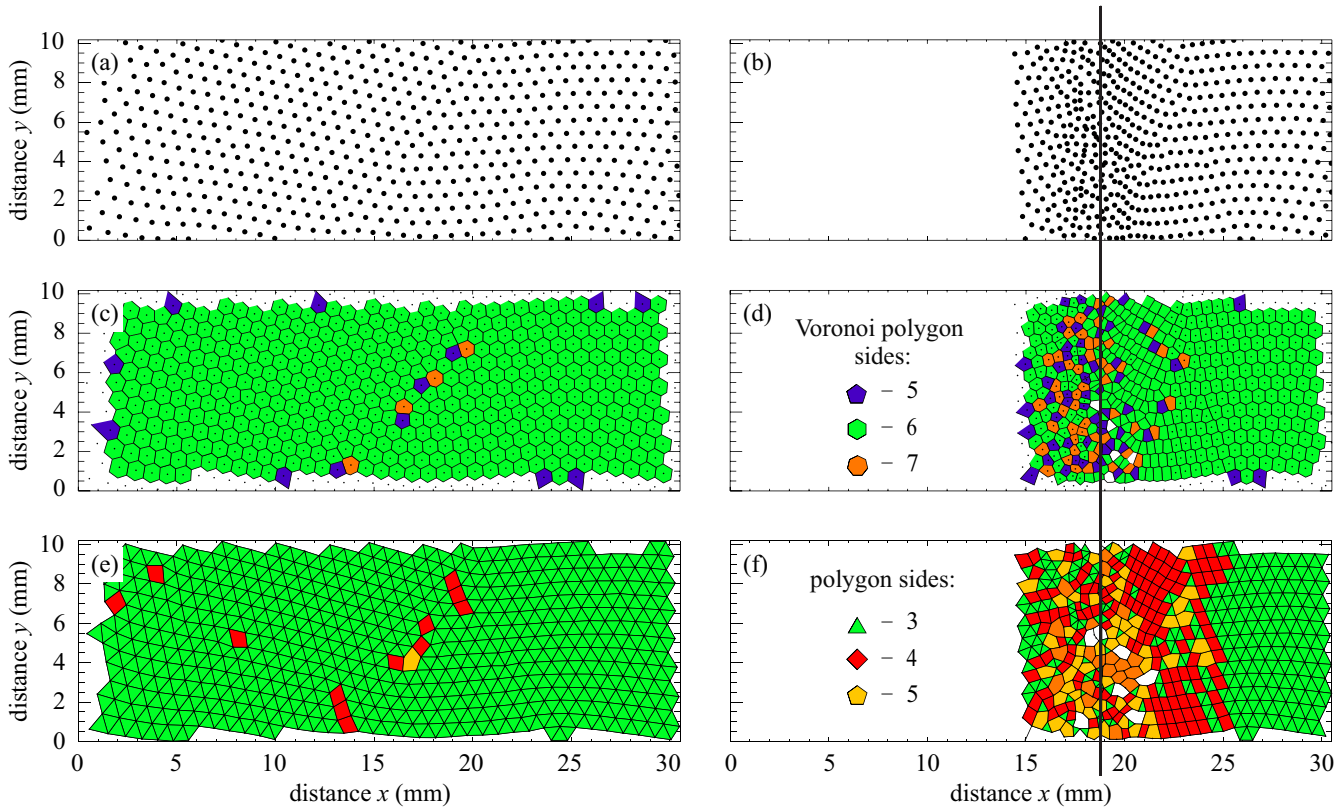

FIG. 1. Microscopic structure of the 2D layer of microparticles. Particle positions are shown in panels (a) and (b). In the region of interest (ROI) shown here, the exciter wire moved from left to right. The left column is for an early time when the shock had not yet disturbed the ROI, while the right column at 248 ms later had a shock at the location marked with a black line. Voronoi diagrams in panels (c) and (d), and polygons in panels (e) and (f), reveal regions of order and disorder differently. In Voronoi diagrams, a topological defect corresponds to a non-sixfold coordination, while in a polygon construction map, a geometric defect is recognized as a polygon with four or more sides. The polygon construction map sensitively reveals a nonhexagonal arrangement, on the front side of the shock, where there is an abundance of quadrilateral polygons. These data are from a run, corresponding to Fig. 2 in Ref. [9], with speeds of 101.6 mm/s for the exciter and 117.5 mm/s for the shock. In the postshock for this run, the crystal has clearly melted, as indicated by the considerable disorder that is typical of a liquid. The snapshots shown here are specific frames from videos r_A.mp4, v_A.mp4, and g_A.mp4 in the Supplemental Material [32].

videos of the particle positions, Voronoi cells, and Glaser-Clark polygons; these videos are for six experimental runs, including the three runs shown in the snapshots of Figs. 1–3.

The shock has a peak density, at a location marked as a solid line in Figs. 1–3. We refer to that point as the shock position. Immediately in front of the shock position there is a localized high-gradient region, which we analyzed in Ref. [14] to quantify the finite thickness of the shock.

Our chief result is the finding that microstructures rearrange from hexagonal to quadrilateral near the shock, and especially on its front side. Such a quadrilateral structure is seen in Figs. 1–3 and in the videos in the Supplemental Material [32].

The quadrilateral structure is most often observed at locations on the front side of the shock. This quadrilateral region is located between what we call the preshock and the postshock. The preshock region is located at the largest value of the x coordinate. Here, we distinguish the quadrilateral region and the postshock not according to a numerical criterion but by our qualitative observation of the type of disorder seen in the local microstructure. In our experiment, the quadrilateral region typically had a thickness of about 4 mm.

An additional result is a considerable increase in defects in the postshock region. Following the previous literature, the defect fraction is quantified as the number of non-sixfold Voronoi cells divided by the total number of cells within a given region. Values of this defect fraction are printed on each video frame in the Voronoi-map movies in the Supplemental Material [32], and they are summarized in Table I. In all six runs the defect fraction behind the shock was mainly greater than the 25% level considered to be a liquid in the melting experiment of Ref. [40], verifying that melting did occur in all six runs. For comparison, the defect fraction was only 3% to 14% in the unperturbed condition, before the shock arrived in the ROI.

Glaser-Clark polygon maps are more useful than Voronoi analysis, for the purpose of identifying a quadrilateral structure. A limitation of Voronoi polygons is that they are most likely to have five, six, or seven sides, while very seldomly do they have four sides. The Voronoi mapping struggles to map onto a quadrilateral arrangement of particles, doing so only by making some of the polygon sides extremely short, as can be seen in the magnified view in Fig. 4. The Voronoi cells in Fig. 4(a) might at a glance appear to be quadrilateral, but a

TABLE I. Runs from the experiment of Ref. [9]. The defect fraction is calculated as the number of non-sixfold Voronoi cells divided by the total number of Voronoi cells, in a spatial region that includes of all the particles located 1.25 mm or more behind the point of highest density. Values for this defect fraction are reported as a typical range over the course of time; values for specific video frames are printed in the videos in the Supplemental Material [32]. The unperturbed conditions were obtained early in each run before the microparticles in the ROI were disturbed noticeably, while the conditions behind the shock are for video frames when the point of maximum density had reached at least the farthest 70% of the ROI. The exciter speed and Mach number data are the same as those in Fig. 5 of Ref. [9]. For comparison, we note that Feng *et al*. [40] found in their melting experiment that a liquid typically has a defect fraction greater than approximately 0.20.

Run	Exciter speed (mm/s)	Exciter speed/$a\omega_p$	Exciter Mach number	Defect fraction: Unperturbed	Defect fraction: Behind the shock
A	101.6	4.9	5.36	0.05–0.08	0.51–0.62
B	101.6	4.9	5.36	0.02–0.06	0.54–0.72
C	88.9	4.3	4.69	0.07–0.08	0.32–0.60
D	76.2	3.7	4.02	0.05–0.08	0.45–0.62
E	63.5	3.1	3.35	0.03–0.10	0.34–0.50
F	50.8	2.5	2.68	0.10–0.14	0.35–0.51

close inspection shows that they are mostly six-sided, along with some that are five- or seven-sided, but always with some sides that are extremely short.

IV. DISCUSSION

An analysis method that had not previously been applied to 2D shocks allowed us to find, on the front side of a shock, a hexagonal microstructure rearranged into a quadrilateral microstructure that had not been reported before. This analysis method was the Glaser-Clark polygon analysis, which we compared to the more common Voronoi analysis method. The data that we analyzed were from runs of the 2D dusty plasma shock experiment of Ref. [9]. An advantage of the Glaser-Clark polygons that we found was that it easily reveals

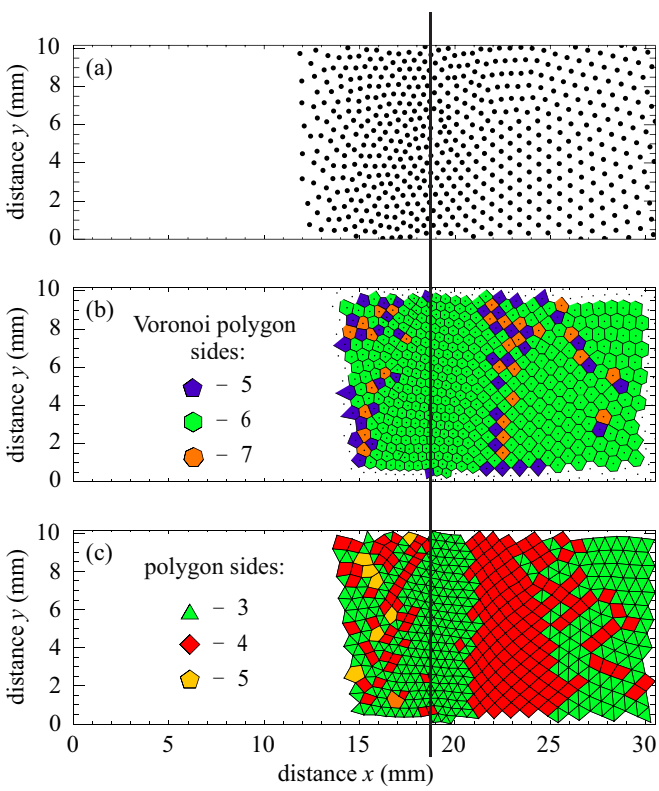

FIG. 2. Microscopic structure, with data from a run with a slower speed of 88.9 mm/s for the exciter and 110.5 mm/s for the shock. In this run, there was again an abundance of quadrilaterals on the front side of the shock. In the postshock, the rearrangement relaxed more nearly toward a hexagonal lattice, meaning that there is no strong indication of melting. These snapshots are from videos r_C.mp4, v_C.mp4, and g_C.mp4 in the Supplemental Material [32].

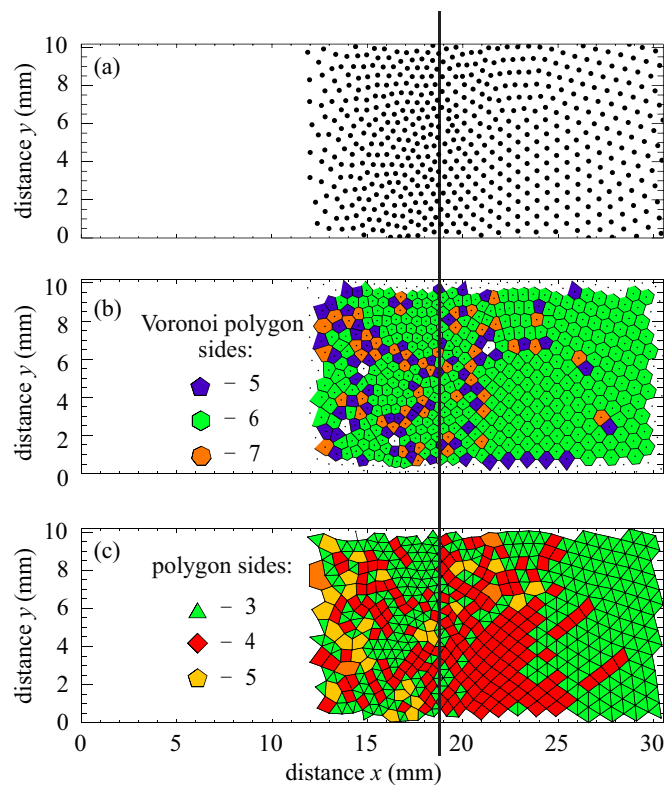

FIG. 3. Microscopic structure, with data from a run with an even slower speed of 63.5 mm/s for the exciter and 83.8 mm/s for the shock. As in the other runs, quadrilaterals were abundant on the front side of the shock. The disorder in the postshock was more than that in Fig. 2, and almost as much as that in Fig. 1. We note that the speed of the shock, by itself, does not determine the outcome of whether the postshock is melted. These snapshots are from videos r_E.mp4, v_E.mp4, and g_E.mp4 in the Supplemental Material [32].

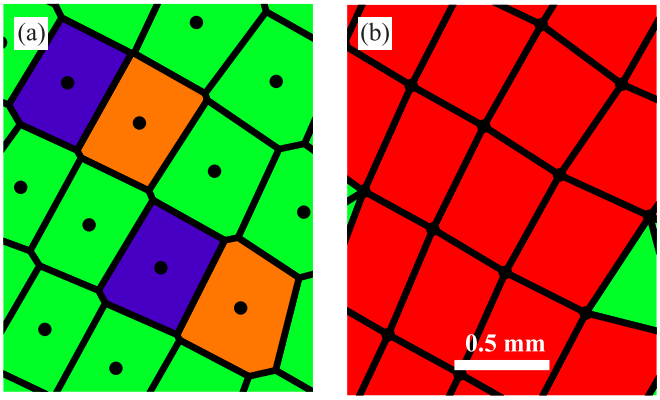

FIG. 4. Magnification of Fig. 1, showing details in the microscopic structure at a location near $x = 21.5$ mm and $y = 7.7$ mm. Under the compressed condition near the shock, sixfold Voronoi cells (a) have sides that are very unequal in length, with some sides that are so short that it is not very meaningful to say that the coordination is sixfold. Under these nonequilibrium conditions, we find that the polygons in panel (b) better indicate the nearly rectangular arrangement of particles.

quadrilateral structures, while Voronoi polygons are not well suited for this purpose, as they are mostly five-, six- or seven-sided. Another advantage of Glaser-Clark polygon analysis is that a gradually increasing disorder in the microstructure is quantified by a gradual increase in the number of sides of the polygons [27].

The extended quadrilateral structure that we observed is generally not otherwise seen in a single layer of identical particles interacting isotropically. A triangular lattice with sixfold symmetry is the only stable lattice in this case. As an example, a square lattice configuration is not stable in equilibrium, as the total interparticle potential energy can be diminished by alternate rows slipping so that the lattice becomes triangular.

We believe that it may be significant that the quadrilateral structure was observed at locations where the conditions were strongly nonequilibrium. Shocks are certainly nonequilibrium, especially near the location of greatest compression, and it was near this location where we observed quadrilateral structures.

In a Voronoi map, the emergence of a defect from a defect-free region requires a change in the nearest neighbors of a particle. We find that this does not generally occur until after the shock has passed. Figures 1 to 3 reveal that, in the vicinity of the shock front, the lattice is merely horizontally compressed, without a topological rearrangement of nearest neighbors. This compression, without a change of nearest neighbors, is consistent with the emergence of quadrilateral polygons without increasing defects in the Voronoi map. This compression introduces a substantial strain energy, making the compressed region unstable. Only after the shock passes is there time for this unstable condition to result in a melted, disordered microstructure, through topological rearrangements.

This finding suggests further work studying nonequilibrium conditions in greater detail using the polygon analysis. Beyond this, Glaser-Clark polygon analysis could be explored for additional applications, such as precise measurements of shock width using the higher sensitivity of the method as compared with the method based on characterizing the coarse-grained density profile [14].

There are previous dusty plasma experiments, done under nearly 2D conditions, where quadrilateral microstructures have been observed under steady conditions. These experiments, which did not involve shocks, fall in two categories: those with an externally imposed pattern and those with out-of-plane particles. An externally imposed pattern was applied to a magnetized plasma, when a metal mesh with a 0.56-mm square pattern was inserted as a boundary in the Magnetized Dusty Plasma Experiment (MDPX) device [41–43], causing the same square pattern to appear in the arrangement of dust particles. This effect can be attributed to the external influence of the mesh, which persisted throughout the volume due to the magnetization of electrons and ions. For unmagnetized plasmas, we note two experiments that involved out-of-plane positions of some particles. In both of these experiments authors attributed their observations of quadrilateral microstructures to nonisotropic interparticle interactions: ion wakes beneath the upper particles that disturbed the lower particles. In 2019, Huang *et al.* [23] reported an experiment with a binary mixture, and they observed distinctive regions with fourfold symmetry rather than the sixfold symmetry that is common in a 2D crystal with monodisperse microparticles. In 2022, Singh *et al.* [25] used monodisperse particles and attained a quadrilateral microstructure by causing the microparticle layer to buckle, meaning that the horizontal repulsion between microparticles overcame the confining forces in the vertical direction.

Shocks in a 2D layer have been studied previously in dusty plasma experiments [5,44] and in molecular dynamics simulations [7,10,12,13,15,16,45]. The latter mimicked dusty plasmas by tracking the motion of identical particles interacting with a repulsive Yukawa potential, but without capturing all the effects in the experiment such as out-of-plane displacements and finite dispersion of particle size. In none of the experiment or simulation papers that we reviewed did the authors comment upon a quadrilateral microstructure. Nevertheless, we see indications of quadrilateral shapes in both the experiment in Fig. 1 of Ref. [44] and in Fig. 1(a) of the Supplemental Material of Ref. [45]. These quadrilateral structures appear in the vicinity of the peak density of a blast-wave shock in the experiment of Ref. [44] but on the front side of shock in the simulation of a steadily-driven shock in Ref. [45].

Besides our primary result of finding a quadrilateral microstructure, we also discussed the production of defects and melting. The defect fraction was considerably higher behind the shock as compared to the unperturbed condition in six runs that were analyzed. In fact, behind the shock the defect fraction was generally so high as to be consistent with melting. We did not observe a distinctive trend for the defect fraction to increase with exciter speed. We note that all of the exciter speeds in the experiment were at Mach numbers considerably higher than those in recent simulations (see Refs. [13] and [12]) where the onset of shock-induced melting was studied. Some factors that could affect the defect fraction behind the shock include the presence, in the earlier unperturbed struc-

ture, of domain walls and other defects, which vary from one experimental run to another. One more factor that can vary from one run to another is the direction of the lattice alignment in the unperturbed condition. These conditions for shocks in a solid differ from those of gasdynamic shocks, where the unshocked region has no microscopic structure or favored directions.

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